

L2.4 Limits at Infinity

§3.4 — where the asymptote rules come from

The three asymptote rules

You already know the left column — it is precalc. The right column is today.

The rule	Where it comes from
① BOT > TOP HA $y = 0$ e.g. $\frac{x^2 + x}{x^3}$	
② BOT = TOP HA at the ratio of leading coefficients e.g. $\frac{3x^2 - 1}{4x^2}$	
③ TOP > BOT no HA; SA by polynomial long division e.g. $\frac{4x^2 - 4}{3x - 6}$	

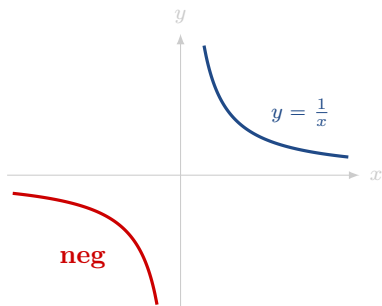
THE LIMIT AT INFINITY THEOREM — all three rules run on this one

$$\lim_{x \rightarrow \infty} \frac{a}{x^r} = 0 \quad a \text{ real, } r \text{ a positive rational}$$

Also true as $x \rightarrow -\infty$, whenever x^r is defined there.

“divide top and bottom by the highest power of x in the BOTTOM, then kill every term that turned into a over x to a power”

$\sqrt{x^2} = |x|$ — the sign on the way to $-\infty$



A square root is never negative, so $\sqrt{x^2}$ is not x — it is $|x|$.

$$x < 0 \implies \frac{1}{x} = -\frac{1}{\sqrt{x^2}}$$

“on the way to $-\infty$ the $\frac{1}{x}$ is negative, and the radical is not, so a minus sign has to appear when you move it inside”

SLANT ASYMPTOTE — and why the line is there

Long division on $\frac{4x^2 - 4}{3x - 6}$ gives

$$\frac{4x^2 - 4}{3x - 6} = \underbrace{\frac{4}{3}x + \frac{8}{3}}_{\rightarrow +\infty} + \underbrace{\frac{12}{3x - 6}}_{\rightarrow 0} \quad SA = \frac{4}{3}x + \frac{8}{3}$$

“the remainder dying is the whole reason the line is an asymptote — the limit is $+\infty$ and the answer is still a line”

$\infty - \infty$ — rewrite first

∞ is not a number, so the limit laws do not apply to it. Rewrite, *then* take the limit. For a difference of two roots, or a root minus a polynomial, multiply by the conjugate over itself — you have done conjugates before.

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x) = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2 + 1} + x} = 0$$

NOTATION

HA horizontal asymptote **VA** vertical asymptote **SA** slant asymptote

Z@ zero at **@** at **TOP / BOT** the degrees, top and bottom

WATCH OUT

✗ $\infty - \infty = 0 \rightarrow \infty$ is not a number — rewrite, then take the limit

✗ divide by the highest power on top $\rightarrow$ by the highest power in the BOTTOM

✗ $\sqrt{x^2} = x \rightarrow \sqrt{x^2} = |x|$, and on the left that is $-x$

✗ a graph has one horizontal asymptote $\rightarrow$ it can have two — one per side, as $\frac{\sqrt{2x^2+1}}{3x-5}$ does

✗ the graph can't cross a horizontal asymptote $\rightarrow$ it can. An HA is end behaviour, not a barrier

✗ no HA means no asymptote $\rightarrow$ TOP one degree higher $\Rightarrow$ SA, by long division

✗ TOP $>$ BOT always gives a slant asymptote $\rightarrow$ only when the degrees differ by exactly one

✗ l'Hôpital gives me the asymptote $\rightarrow$ it gives the limit. The division gives the ratio of leading coefficients, which is the line